

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

6th Semester of B. Pharm. Examination

University Theory Examination April/May 2017

PH322 Phytochemistry- II

Date: 15.05.2017 , Monday

Time: 10.00 a.m. to 01.00 p.m.

Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat labeled sketches wherever necessary.

SECTION - I

- Q 1** Attempt all questions. Each question is of one mark. **20**
1. The biological source of cinnamon:
[A] *Cinnamomum zylenticum*
[B] *Cinnamomum blum*
[C] *Foeniculam vulgare*
[D] *Cinnmomum indicum*
 2. Peppermint oil contains important terpenoid:
[A] Geraniol
[B] Menthol
[C] Citral
[D] Eugenol
 3. Beta – ionone is starting material for synthesis of _____
[A] Carotenoid
[B] Vitamin B
[C] Vitamin A
[D] Lycopene
 4. Monoterpenes contains _____ isoprene unit
[A] 1
[B] 3
[C] 4
[D] 2
 5. _____ is made up of isoprene units
[A] Isoprenoids
[B] Phytones
[C] Flavonoids
[D] Terpenoids
 6. Flavonoids are considered as one of _____ type of compounds
[A] Acidic
[B] Phenolic
[C] Phenyl Isoprenoidal
[D] Phenyl butyl
 7. Central conjugated chain in carotenoid is joined by _____ condensation
[A] Head to Tail
[B] Tail to Tail
[C] Head to Head
[D] Tail to Head

8. Oxygenated carotenoid is termed as _____
 [A] Xanthophylls
 [B] Carotenes
 [C] Lycopene
 [D] Vitamin A
9. One of the following is the commercial source of Pinene
 [A] Fennel
 [B] Turpentine oil
 [C] Citrus oil
 [D] Clove Oil
10. Abietic acid is _____ type of terpenoid
 [A] Sesquiterpenoid
 [B] Triterpenoid
 [C] Monoterpene
 [D] Tetraterpoid
11. Number of unsaturation may be determined using _____
 [A] Iodination
 [B] Oxidation
 [C] Reduction
 [D] Addition reaction
12. Beta carotene on distillation yielded ____ + ____ + ____
 [A] Toluene+ m- xylene + 2,6- dimethyl naphthalene
 [B] Benzene+ m- cresol + naphthalene
 [C] Phenol +m - xylene + 2,6- dimethyl naphthalene
 [D] Abietic Acid
13. How many ionone molecules are obtained from retinol
 [A] 3
 [B] 4
 [C] 1
 [D] 2
14. One of the following is the most preferred analytical instrument to analyze perfume
 [A] High Performance Liquid Chromatography
 [B] Titration
 [C] UV Spectrometry
 [D] Gas Chromatography
15. Concretes are obtained through _____
 [A] Maceration
 [B] Percolation
 [C] Decoction
 [D] Dry Distillation
16. One of following is used to make ice cream blue in colour
 [A] Caramel
 [B] Malvidine
 [C] Bixin
 [D] Indigo
17. Oil from amber is obtained through _____
 [A] Maceration
 [B] Percolation
 [C] Dry Distillation
 [D] Decoction
18. _____ is not used as flavouring agent
 [A] Cardamom
 [B] Lemon
 [C] Rose
 [D] Jasmin
19. Immediately, after spraying the perfume, we have smell of _____
 [A] Top Note

[B] Middle Note

[C] Base Note

[D] All together

20. _____ is responsible for color production for a colorant

[A] Heterocyclic System

[B] Conjugated unsaturation

[C] Unsaturation

[D] Phenyl Ring

SECTION – II

Q 2 Attempt any TWO of the following

A Prove following: 05

1. Lemonene is cyclic compound contains two olefinic linkages

2. Menthol contains secondary alcoholic hydroxyl group

B Prove following: 05

1. One ring in alpha- pinene is six membered ring

2. Alpha- pinene contains one unsaturation

C Apply chemical approach to prove chemical structure of Citral 05

Q 3 Attempt any TWO of the following

A Prepare a note on chemical structure of Flavonoids 05

B Write a note on Beta Carotene with special reference to isolation, chemical structure and uses. 05

C Write important chemical reactions involved in structure elucidation of aglycone of Rutin 05

SECTION – III

Q 4 Attempt any FOUR of the following

A Prepare a note on Marine Sources as potential source of New Drugs 05

B Classify Natural Coloring Agents with examples 05

C Write a note on Herbal Sweetening Agents 05

D Write a note on advantages and disadvantages of natural coloring agents 05

E Biological source, family uses and chemical constituent responsible of at least 5 (Five) natural products, used as source of volatile oil to formulate perfume 05

F Write a detailed note on Preparation of Perfume using natural flavorants 05

Q 5 Attempt any FOUR of the following

A Classify terpenoids on the basis of number of isoprene units giving examples in each class. 05

B Write the chemistry of any one bicyclic monoterpenoid. 05

C Write a note on Retinol 05

D Describe Isoprene rule with suitable example 05

E How will you ascertain functional nature of oxygen in terpenoidal compounds? 05

F Write a short note on: 05

1. Kuhn Roth Oxidation

2. Biogenesis of carotenoids (Mention important steps)